

DD: FUL-07 — HomePass Lifecycle Fulfillment

Developer implementation guide: DD_FUL-07-HomePass_Lifecycle_Fulfillment-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract
DD_FUL-07-HomePass_Lifecycle_Fulfillment-v1.0.md.

Verdict: ■ New | **Wave:** 1 | **Group III:** Fulfillment | **Version:** v1.0 Owns the HomePass-side lifecycle of subscription bindings. When a subscription activates at a HomePass: increments the HomePass's `activeTerminationPoints` and captures the last-mile port binding if `captureMode = AT_INSTALLATION`. When a subscription terminates (or relocates away, migrates to another HomePass, etc.): decrements the counter and releases the port binding. Called synchronously by SUB-WF-ACTIVATE-01 satellites (via the framework's `ful-call-claim-homepass` worker) and by SUB-WF-TERMINATE-01 / SUB-WF-RELOCATION-01 / SUB-WF-MIGRATION-01 satellites (via `ful-call-release-homepass`). Companion to **FUL-03** (network plane activation), **FUL-04** (restriction + non-payment suspension), and **FUL-05** (network plane termination). FUL-07 is the HomePass-plane counterpart — different concern, same architectural pattern.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/06_fulfillment/DD_FUL-07-HomePass_Lifecycle_Fulfillment-v1.0.md
Version	v1.0
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- GET /api/homepass-bindings/{id}
- GET /api/homepass-bindings?homepassId=hp_01HZ_KAREN_017&status=OPEN
- GET /api/homepass-bindings?subscriptionId=sub_01HZ_KAMAU_FIBER
- GET /api/homepass-fulfillment/{id}

- GET /api/homepasses/{id}
- PATCH /api/homepasses/hp_01HZ_DAKAR_012/network-path/append
- PATCH /api/homepasses/{id}/network-path/append
- PATCH /api/homepasses/{id}/network-path/ports
- POST /api/homepass-claim/execute
- POST /api/homepass-release/execute

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- ful-call-claim-homepass
- ful-call-release-homepass
- fulfillment-completed
- fulfillment-failed
- kafka-emit
- manual-recovery-decision

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- HomePassActiveTerminationPointsChanged
- HomePassFulfillmentFailed
- HomePassSubscriptionClaimed
- HomePassSubscriptionEnded

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.

5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 1 | **Group III:** Fulfillment | **Version:** v1.0

Owns the HomePass-side lifecycle of subscription bindings. When a subscription activates at a HomePass: increments the HomePass's `activeTerminationPoints` and captures the last-mile port binding if `captureMode = AT_INSTALLATION`. When a subscription terminates (or relocates away, migrates to another HomePass, etc.): decrements the counter and releases the port binding. Called synchronously by SUB-WF-ACTIVATE-01 satellites (via the framework's `ful-call-claim-homepass` worker) and by SUB-WF-TERMINATE-01 / SUB-WF-RELOCATION-01 / SUB-WF-MIGRATION-01 satellites (via `ful-call-release-homepass`).

Companion to **FUL-03** (network plane activation), **FUL-04** (restriction + non-payment suspension), and **FUL-05** (network plane termination). FUL-07 is the HomePass-plane counterpart — different concern, same architectural pattern.

Glossary

Term	Meaning
HomePass claim	The increment of <code>activeTerminationPoints</code> (and, for <code>AT_INSTALLATION</code> HomePasses, the port-binding capture) when a subscription activates at a HomePass.
HomePass release	The decrement of <code>activeTerminationPoints</code> (and, for <code>AT_INSTALLATION</code> HomePasses, the port-binding teardown) when a subscription terminates / relocates / migrates away from a HomePass.
<code>activeTerminationPoints</code>	Per RLM-CFG-01 R-H-9, the non-negative count of signal-provisioned outlets at a HomePass. Workshop-defined as the basis for contractor payment per active termination point certified. FUL-07 maintains the count in its own binding table; RLM-CFG-01's column is a denormalized read (kept in sync via Kafka event).
HomePass-subscription binding	The runtime record that subscription S is active at HomePass H. One row per active subscription per HomePass in <code>homepass_subscription_binding</code> . Lifecycle: created at claim, soft-closed at release.

Term	Meaning
captureMode	Per RLM-CFG-01 R-H-10, one of PRE_INSTALLATION (full path with ports known at HomePass creation; typical KE) or AT_INSTALLATION (only upstream nodes known; port + last-mile filled by FUL-07 at claim; typical TZ-style fiber-on-customer-install). FUL-07's behavior diverges based on this value.
Port binding	The specific (nodeCode, port) assignment in the HomePass's network_path for a service-management node. For AT_INSTALLATION, FUL-07 captures these at claim time. For PRE_INSTALLATION, the ports were captured at HomePass creation; FUL-07 does not re-capture.
Fulfillment attempt	One end-to-end execution of FUL-07 for a given (subscriptionId, fulfillmentCorrelationId). Idempotency-scoped on this tuple. Recorded in homepass_lifecycle_ledger.

A — Target

1. What this process is about

SUB-WF-ACTIVATE-01 reaches the callFulfillment step in its saga, after FUL-03 has provisioned the network. The saga's ful-call-claim-homepass worker calls FUL-07 with intent=CLAIM. FUL-07 then:

For CLAIM (subscription activating at a HomePass):

1. Reads the HomePass via RLM-CFG-01 (captureMode, networkPath).
2. Inserts a homepass_subscription_binding row tying the subscription to the HomePass.
3. If captureMode = AT_INSTALLATION AND caller supplied portBinding updates: calls PATCH /api/homepasses/{id}/network-path/append (per RLM-CFG-01) to fill in the last-mile node(s) + port(s).
4. Recomputes activeTerminationPoints for the HomePass (count of OPEN bindings).
5. Emits HomePassActiveTerminationPointsChanged event (RLM-CFG-01 consumes to update its denormalized counter column).
6. Emits HomePassSubscriptionClaimed event.
7. Returns homepassClaimId, incrementedActiveTerminationPoints, terminal state.

For RELEASE (subscription terminating / relocating / migrating away):

1. Reads the HomePass + the open homepass_subscription_binding row for this subscription.
2. If no open binding exists for this subscription, returns NO_OPEN_BINDING (informational; non-failing — the subscription may never have claimed, e.g., terminated before activation).
3. If captureMode = AT_INSTALLATION AND releasePortBinding = true: clears the port(s) on the network_path via PATCH /api/homepasses/{id}/network-path/ports setting port=null for the released subscription's last-mile node(s).
4. Soft-closes the binding row (sets released_at, release_reason).
5. Recomputes activeTerminationPoints.
6. Emits HomePassActiveTerminationPointsChanged event.
7. Emits HomePassSubscriptionEnded event.
8. Returns homepassReleaseId, decrementedActiveTerminationPoints, terminal state.

Audit: Every attempt produces a homepass_lifecycle_ledger row with start/end timestamps, terminal state, failure reason if any. Indefinite retention.

Return to caller: The framework worker's external task complete fires, returning the outputs to the BPMN.

Flow overview — CLAIM at activation

```

SUB-WF-ACTIVATE-01-FLOW-WIK reaches ful-call-claim-homepass worker
▼
POST /api/homepass-claim/execute
{ subscriptionId, customerId, operatorCode, homepassId,
  claimReason: "SUBSCRIPTION_ACTIVATED",
  portBindings?: [{nodeCode, port}, ...],    // only if HomePass is AT_INSTALLATION

```

```

    fulfillmentCorrelationId, callerWorkflowKind, callerWorkflowRefId }
    ▼
FUL-07 reads HomePass → inserts binding row → patches network_path if AT_INSTALLATION
→ recomputes count → emits Kafka events → returns claimId
    ▼
{ homepassClaimId, terminalState: "CLAIMED",
  incrementedActiveTerminationPoints: 5,    // count after increment
  capturedPortBindings: [...] }

```

Flow overview — RELEASE at termination

```

SUB-WF-TERMINATE-01-FLOW-WIK reaches ful-call-release-homepass worker
    ▼
POST /api/homepass-release/execute
{ subscriptionId, customerId, operatorCode, homepassId,
  releaseReason: "SUBSCRIPTION_TERMINATED",
  releasePortBinding: true,                // free the port for re-use
  fulfillmentCorrelationId, callerWorkflowKind, callerWorkflowRefId }
    ▼
FUL-07 soft-closes binding → clears port(s) if AT_INSTALLATION
→ recomputes count → emits events → returns releaseId
    ▼
{ homepassReleaseId, terminalState: "RELEASED",
  decrementedActiveTerminationPoints: 4,
  releasedPortBindings: [...] }

```

Out of scope

- **Network provisioning / deprovisioning** — owned by FUL-03 + FUL-05. FUL-07 operates on the HomePass side only.
- **HomePass status transitions** (NPL → SUR → ROE → BLD → RFS → WAI → ACT) — owned by RLM-CFG-01 maker-checker workflow. FUL-07 doesn't change HomePass status; it only updates activeTerminationPoints (derived) and (for AT_INSTALLATION) the per-port assignment within the network_path.
- **HomePass CRUD** — owned by RLM-CFG-01.
- **HomePass-level address / franchise / techregion changes** — owned by RLM-CFG-01.
- **Equipment lifecycle** — owned by OSR-INSTANCE-01. The ONT serial that ends up in the network_path's termination node is an equipment-instance reference; FUL-07 receives it as input from the caller (the workflow knows the equipment binding via OSR-INSTANCE-01) and writes it into the network_path; it does not manage equipment lifecycle.
- **Counter materialization on RLM-CFG-01's homepass.active_termination_points column** — RLM-CFG-01 owns the column; updates from the HomePassActiveTerminationPointsChanged Kafka event handler within RLM-CFG-01.

2. Business rules

Rules grouped: EN (entry / envelope), CL (claim-specific), RL (release-specific), LE (ledger), EV (event emission), CO (compensation).

Rule group EN — Entry + envelope

Rule	Description
R-FUL-07-EN-1	FUL-07 exposes two endpoints: POST /api/homepass-claim/execute (called by ful-call-claim-homepass worker) and POST /api/homepass-release/execute (called by ful-call-release-homepass worker).
R-FUL-07-EN-2	Both endpoints require: operatorCode, subscriptionId, customerId, homepassId, fulfillmentCorrelationId, callerWorkflowKind, callerWorkflowRefId. CLAIM additionally requires claimReason and optionally portBindings[]. RELEASE additionally requires releaseReason and optionally releasePortBinding (default false).
R-FUL-07-EN-3	Idempotency on (subscriptionId, fulfillmentCorrelationId). Re-submission returns the prior fulfillment outcome unchanged.
R-FUL-07-EN-4	The HomePass referenced by homepassId MUST exist in RLM-CFG-01. Otherwise return 404 HOMEPASS_NOT_FOUND.
R-FUL-07-EN-5	operatorCode MUST match the HomePass's franchiseRefs[] franchise operator. Otherwise return 403 OPERATOR_MISMATCH.

Rule group CL — Claim

Rule	Description
R-FUL-07-CL-1	A claim MUST NOT already have an open binding for (subscriptionId, homepassId). If one exists, return 409 BINDING_ALREADY_OPEN with the existing binding_id.
R-FUL-07-CL-2	If HomePass's captureMode = AT_INSTALLATION AND portBindings[] is supplied in the request: FUL-07 calls PATCH /api/homepasses/{id}/network-path/append (or /ports for in-place port-only updates) per RLM-CFG-01 to fill in the captured port assignments. On RLM-CFG-01 PATCH failure → fulfillment fails with PORT_BINDING_PATCH_FAILED; the binding row is NOT inserted.
R-FUL-07-CL-3	If HomePass's captureMode = PRE_INSTALLATION AND portBindings[] is supplied: returns 400 PORT_BINDING_NOT_APPLICABLE (the ports were captured at HomePass creation; FUL-07 does not re-capture).
R-FUL-07-CL-4	If HomePass's captureMode = AT_INSTALLATION AND portBindings[] is NOT supplied: FUL-07 still proceeds with the claim (port capture is optional at claim; can be filled later via direct RLM-CFG-01 PATCH). The homepass_subscription_binding.port_capture_pending flag is set to true.
R-FUL-07-CL-5	Binding row inserted with binding_status = OPEN, claimed_at = NOW(), claim_reason = caller-supplied value. The row's binding_id is returned to the caller as homepassClaimId.
R-FUL-07-CL-6	incrementedActiveTerminationPoints in the response = COUNT(*) FROM homepass_subscription_binding WHERE homepass_id = ? AND binding_status = 'OPEN' after the insert.

Rule group RL — Release

Rule	Description
R-FUL-07-RL-1	Release looks up the OPEN binding row for (subscriptionId, homepassId). If none exists, terminal state is NO_OPEN_BINDING (informational; non-failing per §1; the subscription may have terminated before activation, or already been released). The ledger row records this; no decrement happens; no events emitted.
R-FUL-07-RL-2	If releasePortBinding = true AND HomePass's captureMode = AT_INSTALLATION: FUL-07 calls PATCH /api/homepasses/{id}/network-path/ports per RLM-CFG-01 to set the released subscription's bound port(s) to null (freeing them for re-use by another subscription). The specific nodes to clear are looked up from the binding row's captured_port_bindings JSONB. On RLM-CFG-01 PATCH failure → fulfillment fails with PORT_RELEASE_PATCH_FAILED; the binding row is NOT closed.
R-FUL-07-RL-3	If releasePortBinding = false OR HomePass's captureMode = PRE_INSTALLATION: no port mutation. The ports stay bound on the HomePass's network_path (typical for PRE_INSTALLATION HomePasses where the port is intrinsic to the HomePass).
R-FUL-07-RL-4	Binding row updated: binding_status = CLOSED, released_at = NOW(), release_reason = caller-supplied.
R-FUL-07-RL-5	decrementedActiveTerminationPoints in the response = COUNT(*) FROM homepass_subscription_binding WHERE homepass_id = ? AND binding_status = 'OPEN' after the update.

Rule group LE — Ledger

Rule	Description
R-FUL-07-LE-1	One ledger row per fulfillment attempt in homepass_lifecycle_ledger. Unique on (subscription_id, fulfillment_correlation_id). Carries the envelope fields, intent (CLAIM / RELEASE), terminal state, timestamps.
R-FUL-07-LE-2	Ledger is append-only at the application layer. Updates only happen within the same fulfillment attempt. After terminal state reached, no further updates.
R-FUL-07-LE-3	Retention: indefinite. The ledger participates in the daily BSS↔NMS reconciliation flagged by workshop sources.

Rule group EV — Event emission

Rule	Description	
R-FUL-07-EV-1	On successful CLAIM, emit HomePassSubscriptionClaimed to <code>sophix.fulfillment.homepass.claimed</code> . Carries <code>homepassId</code> , <code>subscriptionId</code> , <code>customerId</code> , <code>operatorCode</code> , <code>claimedAt</code> , <code>claimReason</code> , <code>bindingId</code> , <code>capturedPortBindings</code> .	
R-FUL-07-EV-2	On successful RELEASE, emit HomePassSubscriptionEnded to <code>sophix.fulfillment.homepass.ended</code> . Carries <code>homepassId</code> , <code>subscriptionId</code> , <code>customerId</code> , <code>operatorCode</code> , <code>releasedAt</code> , <code>releaseReason</code> , <code>bindingId</code> , <code>releasedPortBindings</code> .	
R-FUL-07-EV-3	On any change to a HomePass's open-binding count, emit HomePassActiveTerminationPointsChanged to <code>sophix.fulfillment.homepass.atpchanged</code> . Carries <code>homepassId</code> , <code>operatorCode</code> , <code>newActiveTerminationPoints</code> , <code>previousActiveTerminationPoints</code> , <code>changeReason</code> ('CLAIM')	'RELEASE'), <code>triggerSubscriptionId</code> . RLM-CFG-01 consumes to update its denormalized <code>homepass.active_termination_points</code> column.
R-FUL-07-EV-4	Events emitted via kafka-emit worker pattern (transactional outbox) — writes to <code>kafka_outbox</code> in the same DB transaction as the binding row mutation.	

Rule group CO — Compensation

Rule	Description
R-FUL-07-CO-1	Compensation pattern: a CLAIM that fails after partial commit (e.g., port PATCH succeeded but binding INSERT failed, or vice versa) reverts via the inverse — calls <code>PATCH /api/homepasses/{id}/network-path/ports</code> to null the ports OR DELETES the binding row, depending on which step succeeded.
R-FUL-07-CO-2	A RELEASE failure post-binding-close: the binding row stays CLOSED but the port PATCH didn't fire. Recorded in the ledger as <code>RELEASED_PORT_PATCH_FAILED</code> ; manual NOC recovery via the <code>manual-recovery-decision</code> user task.
R-FUL-07-CO-3	A failed CLAIM does NOT trigger compensation against the network plane (FUL-03 already succeeded by the time FUL-07 runs in the ACTIVATE BPMN; the BPMN's own compensation path handles network-plane rollback if needed). FUL-07 only reverts its own writes.

3. Schemas

homepass_subscription_binding

The runtime source of truth for "which subscriptions are active at which HomePasses". `activeTerminationPoints` is `COUNT(*) WHERE binding_status = 'OPEN'`.

Column	Type	Notes
<code>binding_id</code>	TEXT PK	ULID. Example: <code>hsb_01HZ_KAMAU_KAREN17</code>
<code>operator_code</code>	TEXT NOT NULL	WIK / WUG / WTZ / YASSN
<code>subscription_id</code>	TEXT NOT NULL	FK to subscription. Example: <code>sub_01HZ_KAMAU_FIBER</code>
<code>customer_id</code>	TEXT NOT NULL	Denormalized. Example: <code>cust_01HZ_KAMAU</code>
<code>homepass_id</code>	TEXT NOT NULL	FK to homepass. Example: <code>hp_01HZ_KAREN_017</code>
<code>binding_status</code>	TEXT NOT NULL	OPEN (subscription currently active at this HomePass) / CLOSED (released)
<code>claim_reason</code>	TEXT NOT NULL	Caller-supplied reason at claim time. Examples: <code>SUBSCRIPTION_ACTIVATED</code> , <code>SUBSCRIPTION_RELOCATED_TO_HERE</code> , <code>SUBSCRIPTION_MIGRATED_TO_HERE</code>

Column	Type	Notes
release_reason	TEXT NULL	Caller-supplied reason at release time. Examples: SUBSCRIPTION_TERMINATED, SUBSCRIPTION_RELOCATED_AWAY, SUBSCRIPTION_MIGRATED_AWAY. Null while OPEN.
captured_port_bindings	JSONB NULL	For captureMode = AT_INSTALLATION: the port assignments captured at claim. Array of {nodeCode, port} items copied from the request. Used at release time to know which ports to null. Null for PRE_INSTALLATION HomePasses.
port_capture_pending	BOOLEAN NOT NULL DEFAULT FALSE	True if captureMode = AT_INSTALLATION AND no portBindings[] supplied at claim. Cleared when ports later captured via direct RLM-CFG-01 PATCH (out of scope for FUL-07 but observable).
claimed_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()	
released_at	TIMESTAMPTZ NULL	Null while OPEN
claim_correlation_id	TEXT NOT NULL	Caller's fulfillmentCorrelationId at claim
release_correlation_id	TEXT NULL	Caller's fulfillmentCorrelationId at release
caller_workflow_kind_at_claim	TEXT NOT NULL	SUB_WF_ACTIVATE (also future SUB_WF_RELOCATION / SUB_WF_MIGRATION as destination)
caller_workflow_kind_at_release	TEXT NULL	SUB_WF_TERMINATE / SUB_WF_RELOCATION / SUB_WF_MIGRATION (as source)
caller_workflow_ref_id_at_claim	TEXT NOT NULL	
caller_workflow_ref_id_at_release	TEXT NULL	

Indexes:

- UNIQUE (homepass_id, subscription_id) WHERE binding_status = 'OPEN' — enforces R-CL-1
- idx_hsb_homepass_open ON homepass_subscription_binding (homepass_id) WHERE binding_status = 'OPEN' — fast count for activeTerminationPoints
- idx_hsb_subscription ON homepass_subscription_binding (subscription_id, claimed_at DESC) — subscription's binding history
- idx_hsb_operator_status ON homepass_subscription_binding (operator_code, binding_status, claimed_at DESC) — ops dashboards

Sample data:

binding_id	operator	subscription_id	homepass_id	status	claim_reason	release_reason	captured_port_bindings	claimed_at	released_at
hsb_01HZ_KAMAU_KAREN17	WIK	sub_01HZ_KAMAU_FIBER	hp_01HZ_KAREN_017	CLOSED	SUBSCRIPTION_ACTIVATED	SUBSCRIPTION_TERMINATED	NULL (KE PRE_INSTALLATION)	2024-01-15 10:02:22+03	2026-07-20 14:30:21+03
hsb_01HZ_NJERI_LAVI03	WIK	sub_01HZ_NJERI_TRIPLAY	hp_01HZ_LAVI_003	OPEN	SUBSCRIPTION_ACTIVATED	NULL	NULL	2025-08-12 09:15:00+03	NULL
hsb_01HZ_DIOP_DAKAR12	YASSN	sub_01HZ_DIOP_FIBER	hp_01HZ_DAKAR_012	OPEN	SUBSCRIPTION_ACTIVATED	NULL	[{"nodeCode": "OLT-DKR-MED-02", "port": "1/4/7"}, {"nodeCode": "ONT-DIOP", "port": null}]	2025-11-03 14:00:00+00	NULL

Row 1: Kamau's binding closed at termination. Captured port bindings null because KE HomePasses are PRE_INSTALLATION (port captured at HomePass creation, not at customer install).

Row 3: Diop in Senegal. YASSN deployment uses AT_INSTALLATION; the port 1/4/7 on OLT-DKR-MED-02 was captured at FUL-07 claim time and patched into the HomePass network_path. Released_at will be set when Diop terminates.

homepass_lifecycle_ledger

One row per FUL-07 attempt (CLAIM or RELEASE).

Column	Type	Notes
id	TEXT PK	ULID. Example: hll_01HZ_KAMAU_CLAIM
operator_code	TEXT NOT NULL	
subscription_id	TEXT NOT NULL	
customer_id	TEXT NOT NULL	
fulfillment_correlation_id	TEXT NOT NULL	Caller's saga operation ID
caller_workflow_kind	TEXT NOT NULL	SUB_WF_ACTIVATE / SUB_WF_TERMINATE / SUB_WF_RELOCATION / SUB_WF_MIGRATION
caller_workflow_ref_id	TEXT NOT NULL	
homepass_id	TEXT NOT NULL	
intent	TEXT NOT NULL	CLAIM / RELEASE
claim_reason	TEXT NULL	Set for CLAIM only
release_reason	TEXT NULL	Set for RELEASE only
port_bindings_input	JSONB NULL	The portBindings[] from the request (CLAIM) or the bound ports being released (RELEASE)
binding_id	TEXT NULL	The homepass_subscription_binding.binding_id affected. Set when the binding mutation succeeds; null on early failure.
terminal_state	TEXT NOT NULL	CLAIMED / RELEASED / NO_OPEN_BINDING / FAILED_PORT_PATCH / FAILED_BINDING_MUTATION / IN_PROGRESS
failure_reason_code	TEXT NULL	E.g., HOMEPASS_NOT_FOUND, BINDING_ALREADY_OPEN, PORT_BINDING_NOT_APPLICABLE, PORT_BINDING_PATCH_FAILED
failure_reason_detail	TEXT NULL	Free text
previous_active_termination_points	INTEGER NULL	Counter value before this attempt's effect
new_active_termination_points	INTEGER NULL	Counter value after
started_at	TIMESTAMPTZ NOT NULL DEFAULT NOW()	
completed_at	TIMESTAMPTZ NULL	
duration_ms	INTEGER NULL	

Indexes:

- UNIQUE (subscription_id, fulfillment_correlation_id) — idempotency
- idx_hll_homepass ON homepass_lifecycle_ledger (homepass_id, started_at DESC) — homepass-side history
- idx_hll_terminal_state ON homepass_lifecycle_ledger (operator_code, terminal_state, started_at DESC) — ops dashboards
- idx_hll_caller ON homepass_lifecycle_ledger (caller_workflow_kind, caller_workflow_ref_id) — trace from workflow

Sample data:

id	operator	subscription_id	homepass_id	intent	terminal_state	claim_reason	release_reason	prev_atp	new_atp
hll_01HZ_KAMAU_CLAIM	WIK	sub_01HZ_KAMAU_FIBER	hp_01HZ_KAREN_017	CLAIM	CLAIMED	SUBSCRIPTION_ACTIVATED	NULL	4	5
hll_01HZ_KAMAU_REL	WIK	sub_01HZ_KAMAU_FIBER	hp_01HZ_KAREN_017	RELEASE	RELEASED	NULL	SUBSCRIPTION_TERMINATED	5	4
hll_01HZ_NJERI_REL_N00P	WIK	sub_01HZ_NJERI_TRIPLPLAY	hp_01HZ_LAVI_003	RELEASE	NO_OPEN_BINDING	NULL	SUBSCRIPTION_TERMINATED	NULL	NULL
hll_01HZ_DIOP_CLAIM	YASSN	sub_01HZ_DIOP_FIBER	hp_01HZ_DAKAR_012	CLAIM	CLAIMED	SUBSCRIPTION_ACTIVATED	NULL	0	1

Row 3: a release was called against a subscription that had no open binding at the HomePass (e.g., subscription terminated before activation completed). Terminal state NO_OPEN_BINDING — informational, non-failing.

4. API surface

Execute HomePass claim

```
POST /api/homepass-claim/execute HTTP/1.1
Authorization: Bearer <internal-svc-jwt>
Content-Type: application/json

{
  "operatorCode":      "WIK",
  "subscriptionId":    "sub_01HZ_KAMAU_FIBER",
  "customerId":        "cust_01HZ_KAMAU",
  "homepassId":        "hp_01HZ_KAREN_017",
  "claimReason":       "SUBSCRIPTION_ACTIVATED",
  "portBindings":      [],
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_ACT:claim-homepass",
  "callerWorkflowKind": "SUB_WF_ACTIVATE",
  "callerWorkflowRefId": "subop_01HZ_KAMAU_ACT"
}

→ 200 OK
{
  "homepassClaimId":      "hll_01HZ_KAMAU_CLAIM",
  "bindingId":            "hsb_01HZ_KAMAU_KAREN17",
  "terminalState":        "CLAIMED",
  "incrementedActiveTerminationPoints": 5,
  "previousActiveTerminationPoints": 4,
  "capturedPortBindings": [],
  "completedAt":          "2024-01-15T10:02:22.500+03:00",
  "durationMs":            180
}
```

YASSN AT_INSTALLATION example with port capture:

```
POST /api/homepass-claim/execute
{
  "operatorCode":      "YASSN",
  "subscriptionId":    "sub_01HZ_DIOP_FIBER",
  "customerId":        "cust_01HZ_DIOP",
  "homepassId":        "hp_01HZ_DAKAR_012",
  "claimReason":       "SUBSCRIPTION_ACTIVATED",
  "portBindings": [
    { "nodeCode": "OLT-DKR-MED-02", "port": "1/4/7" },
    { "nodeCode": "ONT-DIOP", "port": null }
  ],
  "fulfillmentCorrelationId": "subop_01HZ_DIOP_ACT:claim-homepass",
  "callerWorkflowKind":      "SUB_WF_ACTIVATE",
  "callerWorkflowRefId":     "subop_01HZ_DIOP_ACT"
}

→ 200 OK
{
  "homepassClaimId": "hll_01HZ_DIOP_CLAIM",
  "bindingId":       "hsb_01HZ_DIOP_DAKAR12",
  "terminalState":   "CLAIMED",
  "incrementedActiveTerminationPoints": 1,
  "previousActiveTerminationPoints": 0,
  "capturedPortBindings": [
    { "nodeCode": "OLT-DKR-MED-02", "port": "1/4/7" },
    { "nodeCode": "ONT-DIOP", "port": null }
  ]
}
```

Binding already open:

```
→ 409 Conflict
{
  "code": "BINDING_ALREADY_OPEN",
  "existingBindingId": "hsb_01HZ_KAMAU_KAREN17",
  "claimedAt": "2024-01-15T10:02:22+03:00"
}
```

Port binding not applicable (KE PRE_INSTALLATION + portBindings supplied):

```
→ 400 Bad Request
{
  "code": "PORT_BINDING_NOT_APPLICABLE",
  "homepassCaptureMode": "PRE_INSTALLATION"
}
```

Execute HomePass release

```
POST /api/homepass-release/execute HTTP/1.1
Authorization: Bearer <internal-svc-jwt>
Content-Type: application/json
```

```
{
  "operatorCode":      "WIK",
  "subscriptionId":    "sub_01HZ_KAMAU_FIBER",
  "customerId":        "cust_01HZ_KAMAU",
  "homepassId":        "hp_01HZ_KAREN_017",
  "releaseReason":     "SUBSCRIPTION_TERMINATED",
  "releasePortBinding": false,
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_TERM:release-homepass",
  "callerWorkflowKind": "SUB_WF_TERMINATE",
  "callerWorkflowRefId": "subop_01HZ_KAMAU_TERM"
}
```

→ 200 OK

```
{
  "homepassReleaseId": "hll_01HZ_KAMAU_REL",
  "bindingId":         "hsb_01HZ_KAMAU_KAREN17",
  "terminalState":     "RELEASED",
  "decrementedActiveTerminationPoints": 4,
  "previousActiveTerminationPoints":    5,
  "releasedPortBindings":               [],
  "completedAt":                       "2026-07-20T14:30:21+03:00",
  "durationMs":                        120
}
```

No open binding (terminated before activation):

→ 200 OK

```
{ "homepassReleaseId": "hll_01HZ_NJERI_REL_NOOP",
  "bindingId":         null,
  "terminalState":     "NO_OPEN_BINDING",
  "decrementedActiveTerminationPoints": 7,
  "previousActiveTerminationPoints":    7 }
```

Read a fulfillment

GET /api/homepass-fulfillment/{id} → ledger row

GET /api/homepass-bindings/{id} → binding row

GET /api/homepass-bindings?homepassId=hp_01HZ_KAREN_017&status=OPEN → bindings for one HomePass

GET /api/homepass-bindings?subscriptionId=sub_01HZ_KAMAU_FIBER → bindings for one subscription (history)

5. Kafka events emitted

Topic prefix: `sophix.fulfillment.homepass.*`

Event	Topic	Fires on	Consumers
HomePassSubscriptionClaimed	<code>sophix.fulfillment.homepass.claimed</code>	Successful CLAIM (terminal CLAIMED)	Calling SUB-WF-ACTIVATE BPMN via Camunda message correlator (fulfillment-completed), CVM analytics, EM-04
HomePassSubscriptionEnded	<code>sophix.fulfillment.homepass.ended</code>	Successful RELEASE (terminal RELEASED)	Calling SUB-WF-TERMINATE / SUB-WF-RELOCATION / SUB-WF-MIGRATION BPMN, CVM (churn-by-homepass analytics), EM-04
HomePassActiveTerminationPointsChanged	<code>sophix.fulfillment.homepass.atpchanged</code>	Any CLAIM or RELEASE that changes the count	RLM-CFG-01 (updates its denormalized <code>homepass.active_termination_points</code> column), EM-04 reporting (contractor-payment basis aggregation), monitoring
HomePassFulfillmentFailed	<code>sophix.fulfillment.homepass.failed</code>	Terminal FAILED_PORT_PATCH / FAILED_BINDING_MUTATION	Calling BPMN (fulfillment-failed message), ICN-01 ops alerts

Sample HomePassSubscriptionClaimed:

```
{
  "eventType": "HomePassSubscriptionClaimed",
  "aggregateId": "hsb_01HZ_KAMAU_KAREN17",
  "timestamp": "2024-01-15T10:02:22.500+03:00",
  "payload": {
    "operatorCode": "WIK",
    "homepassClaimId": "hll_01HZ_KAMAU_CLAIM",
    "bindingId": "hsb_01HZ_KAMAU_KAREN17",
    "subscriptionId": "sub_01HZ_KAMAU_FIBER",
    "customerId": "cust_01HZ_KAMAU",
    "homepassId": "hp_01HZ_KAREN_017",
    "claimedAt": "2024-01-15T10:02:22+03:00",
    "claimReason": "SUBSCRIPTION_ACTIVATED",
    "capturedPortBindings": []
  }
}
```

```

    "fulfillmentCorrelationId": "subop_01HZ_KAMAU_ACT:claim-homepass"
  }
}

```

Sample HomePassActiveTerminationPointsChanged:

```

{
  "eventType": "HomePassActiveTerminationPointsChanged",
  "aggregateId": "hp_01HZ_KAREN_017",
  "timestamp": "2024-01-15T10:02:22.500+03:00",
  "payload": {
    "operatorCode": "WIK",
    "homepassId": "hp_01HZ_KAREN_017",
    "previousActiveTerminationPoints": 4,
    "newActiveTerminationPoints": 5,
    "changeReason": "CLAIM",
    "triggerSubscriptionId": "sub_01HZ_KAMAU_FIBER",
    "triggerBindingId": "hsb_01HZ_KAMAU_KAREN17"
  }
}

```

RLM-CFG-01 consumes this event and updates `homepass.active_termination_points` accordingly (own-domain materialization).

6. Worked example — Diop's activation at a YASSN HomePass (AT_INSTALLATION)

YASSN deploys with `captureMode = AT_INSTALLATION` per workshop. Diop is being installed at a HomePass in Dakar; the FUL-02 onboarding captured upstream nodes at HomePass creation but the specific OLT port + ONT serial weren't known. Field technician arrives, plugs in ONT ONT-DIOP, captures the OLT port 1/4/7 on OLT-DKR-MED-02, transmits via the field tablet.

SUB-WF-ACTIVATE-01-FLOW-YASSN reaches the `ful-call-claim-homepass` worker. The worker calls FUL-07:

14:00:00:

```

POST /api/homepass-claim/execute
{ "operatorCode": "YASSN", "subscriptionId": "sub_01HZ_DIOP_FIBER",
  "customerId": "cust_01HZ_DIOP", "homepassId": "hp_01HZ_DAKAR_012",
  "claimReason": "SUBSCRIPTION_ACTIVATED",
  "portBindings": [
    { "nodeCode": "OLT-DKR-MED-02", "port": "1/4/7"},
    { "nodeCode": "ONT-DIOP", "port": null}
  ],
  "fulfillmentCorrelationId": "subop_01HZ_DIOP_ACT:claim-homepass",
  "callerWorkflowKind": "SUB_WF_ACTIVATE",
  "callerWorkflowRefId": "subop_01HZ_DIOP_ACT" }
→ 200 IN_PROGRESS, hll_01HZ_DIOP_CLAIM

```

14:00:00.050: FUL-07 looks up HomePass `hp_01HZ_DAKAR_012` via RLM-CFG-01 read. Returns `captureMode = AT_INSTALLATION`, `network_path` with upstream nodes (HEADEND, splitter) but downstream entries empty.

14:00:00.100: No existing OPEN binding for (`sub_01HZ_DIOP_FIBER`, `hp_01HZ_DAKAR_012`) — clear to proceed.

14:00:00.150: Calls `PATCH /api/homepasses/hp_01HZ_DAKAR_012/network-path/append` per RLM-CFG-01 with body `{nodes: [{type: "OLT", code: "OLT-DKR-MED-02", role: "service_management", port: "1/4/7"}, {type: "ONT", code: "ONT-DIOP", role: "termination", port: null}]}`. RLM-CFG-01 validates OLT-DKR-MED-02 is an ACTIVE `network_node`, appends, re-derives `serviceManagementEndpoints` to include `{data: {nodeCode: "OLT-DKR-MED-02", port: "1/4/7"}}`. Returns 200.

14:00:00.350: Inserts `homepass_subscription_binding` row: `binding_id=hsb_01HZ_DIOP_DAKAR12`, `binding_status=OPEN`, `captured_port_bindings=[{nodeCode:OLT-DKR-MED-02,port:1/4/7},{nodeCode:ONT-DIOP,port:null}]`, `claimed_at=2025-11-03T14:00:00.350+00`.

14:00:00.400: Counts OPEN bindings for `hp_01HZ_DAKAR_012` → 1 (was 0).

14:00:00.450: Writes 2 outbox rows: `HomePassSubscriptionClaimed`, `HomePassActiveTerminationPointsChanged`. Outbox dispatcher publishes within seconds.

14:00:00.500: Returns to caller worker:

```

{ "homepassClaimId": "hll_01HZ_DIOP_CLAIM", "bindingId": "hsb_01HZ_DIOP_DAKAR12",
  "terminalState": "CLAIMED",
  "incrementedActiveTerminationPoints": 1, "previousActiveTerminationPoints": 0,
  "capturedPortBindings": [...], "completedAt": "2025-11-03T14:00:00.500+00",
  "durationMs": 480 }

```

14:00:01+ (downstream):

- RLM-CFG-01 consumes HomePassActiveTerminationPointsChanged, updates hp_01HZ_DAKAR_012.active_termination_points from 0 to 1.
- The SUB-WF-ACTIVATE-01-FLOW-YASSN BPMN's Camunda message correlator catches HomePassSubscriptionClaimed (correlated on fulfillmentCorrelationId) and advances.
- EM-04 indexes the event for monthly contractor-payment aggregation (1 new active termination point this month at this HomePass).

Diop's network is now fully bound to the HomePass: the OLT port is reserved for him, the ONT is provisioned, the network_path describes the complete route.

7. Cross-DD coordination

Module	Direction	Contract
SUB-WF-ACTIVATE-01-FLOW-WIK + future -FLOW- {operator}	Calls FUL-07 via ful-call-claim-homepass worker	Synchronous REST to POST /api/homepass-claim/execute
SUB-WF-TERMINATE-01-FLOW-WIK + future -FLOW- {operator}	Calls FUL-07 via ful-call-release-homepass worker	Synchronous REST to POST /api/homepass-release/execute
SUB-WF-RELOCATION-01 (pending)	Calls FUL-07 release at source HomePass + claim at destination HomePass	Same two endpoints; sequenced in the relocation BPMN
SUB-WF-MIGRATION-01 (pending)	Similar to relocation — release source, claim destination	Same
RLM-CFG-01	FUL-07 reads HomePass (captureMode, networkPath), patches networkPath (AT_INSTALLATION only); RLM-CFG-01 consumes HomePassActiveTerminationPointsChanged to update its denormalized counter	Read: GET /api/homepasses/{id}. Write: PATCH /api/homepasses/{id}/network-path/append, PATCH /api/homepasses/{id}/network-path/ports. Event subscription: sophix.fulfillment.homepass.atpchanged
EM-04	Consumes events	Contractor-payment aggregation (basis = sum of activeTerminationPoints increments per period per franchise)
CVM	Consumes events	Churn-by-HomePass analytics
OSR-INSTANCE-01	Indirect — FUL-07 receives portBindings[] from the caller workflow, which sourced them from the field tech via OSR-INSTANCE-01 equipment-binding capture	No direct call from FUL-07

8. Dependencies

Dependency	Owner	Status
RLM-CFG-01 HomePass (read + PATCH network-path)	network/plm team	✓ v1.0
SUB-WF-FRAMEWORK-01-WORKERS (ful-call-claim-homepass, ful-call-release-homepass contracts)	subscription team	✓ v1.0
SUB-WF-ACTIVATE-01 + FLOW-WIK	subscription team	✓ v1.0
SUB-WF-TERMINATE-01 + FLOW-WIK	subscription team	✓ v1.0
FOUNDATION_DB, FOUNDATION_KAFKA, FOUNDATION_AUTH	infra team	✓

B — Current TZ

Status: To be filled in after codebase cartography review.

Workshop materials note that active termination points are tracked as the basis for contractor payment ("active termination points certified per contractor per period"), and that TZ-style deployments use the AT_INSTALLATION model where field ops captures port details at install. Specific table names, current counter materialization, current handling of port-binding lifecycle vs HomePass lifecycle in the legacy codebase require source-code review to document accurately.

Until that cartography lands, this DD does not assert specific legacy structural claims. The V3 design above stands on its own — RLM-CFG-01 owns the HomePass catalog, FUL-07 owns the subscription↔HomePass binding lifecycle, the two are coordinated via a HomePass-side denormalized counter materialization fed by the HomePassActiveTerminationPointsChanged Kafka event.